THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

032/2B CHEMISTRY 2B

(Actual Practical)

TIME: 2:30 HOURS **YEAR: 2025**

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **all** questions.

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3. Each question carries **twenty-five (25)** marks
4. Non programming calculators may be used.
5. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer sheets provided.
8. You may use the following constants:-

   Atomic masses: H = 1, O = 16, Na= 23, C=12, S =32, Cl = 35.5

   1litre = 1dm3 = 1000cm3

1. You are provided with the following solutions.

C: containing 3_g of acetic acid (CH_3COOH) in $0.5dm^3$ of solution.

D: Containing 1.25g of impure potassium hydroxide in $250cm^3$of solution.

Phenolphthalein indicator (POP)

PROCEDURES:

(i) Fill the burette with solution **C**.

(ii) Use the pipette to transfer $25cm^3$or $20cm^3$of the **D**, into clean and dry conical flask and three (03) drops of POP.

(iii) Titrate the resulting mixture against the solution C from the burette.

(iv) Repeat the procedures (i) to (iii) to obtain three more readings and record the result in a tabular form.

QUESTIONS:

(a) Is the use methyl orange indicator in this experiment suitable as phenolphthalein? Give a reason for your answer.

(b) (i) _____cm^3 of **C** required_____cm^3 of **D** for complete reaction.

(ii) Write a balanced chemical equation for reaction between **C** and **D**.

(c) Showing your procedures clearly /calculate the percentage purity of potassium hydroxide.

2. You are provided with the following

- Solution **ZZ** containing 0.5M sodium thiosulphate ($Na_2S_2O_3$)
- Solution **TT** containing 0.25M nitric acid (HNO_3)
- Distilled water
- Piece of paper marked X
- Stop watch

Procedures

(i) Using measuring cylinder measure $5cm^3$ of solution ZZ and put into a beaker of 100 cm^3

(ii) Measure $5cm^3$ of solution TT put into 100 cm^3 beaker containing solution ZZ and immediately start the stop watch

(iii) Swirl the content in the 100 cm^3 beaker and put the beaker an top of mark X on the piece of paper. Watch the mixture from the above and observe changes.

(iv) Stop the watch when the letter X disappear

(v) Record the time taken for the letter X to disappear

(vi) Repeat procedures (i) to (v) using data shown in table 1

Table 1

Experiment	Vol. Of TT (cm^3)	Vol. Of ZZ (cm^3)	Vol. Of distilled water (cm^3)	Time (s)
1	5	5	0	
2	5	4	1	
3	5	3	2	
4	5	2	3	

Questions

(a) Complete table 1

(b) Write a balanced equation for reaction between T and Z

(c) What substance was provided during the reaction which obscured letter X?

(d) Plot a graph of volume of $Na_2S_2O_3$ solution against time (s)

(e) What conclusion can you draw from this experiment?